

claude-windows / aa329f09-28b7-4acc-b062-98ec7e905abc

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\aa329f09-28b7-4acc-b062-98ec7e905abc\subagents\workflows\wf\_ddd29371-8c2\agent-a448eabd2e47f7d1d.jsonl`
- SHA-256: `f021a53e6ec393e1838f92974bcf2b93c23ecd92bcd17a8196ff743b6d189e7a`
- Source modified: `2026-05-30T03:08:34+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf\_ddd29371-8c2`
- session_id: `aa329f09-28b7-4acc-b062-98ec7e905abc`

Transcript

1. user (2026-05-30T03:03:23.570Z)

Web Searcher: Shuttle/ball trajectory tracking repo licenses (commercial-use viability)

Research question: "RESEARCH GOAL: Identify and eliminate every dependency that could legally or economically hinder monetizing the "badminton-highlight-indexer" project as a HOSTED API SaaS. Business model: the project runs server-side on a cloud GPU VM (NVIDIA); users upload their own badminton videos and receive auto-generated highlight reels. CRITICAL CONSTRAINT: we do NOT distribute the code or any binary to users ? everything stays server-side ? so software that is merely *hosted* on the server (invoked as a separate process, e.g. ffmpeg) is acceptable where distribution-triggered copyleft (GPL/LGPL) would NOT be triggered, BUT network/SaaS copyleft (AGPL §13) and external metered-API terms STILL apply. Preference order: permissively-licensed (Apache-2.0/MIT/BSD) and royalty-free, self-hostable components. Paid commercial licenses (e.g. Ultralytics Enterprise, codec patent licenses) are acceptable ONLY as clearly-flagged fallbacks. Output must be decision-grade with licenses verified against PRIMARY sources (actual GitHub LICENSE files, official pricing/terms pages), because this drives a real business decision ? flag any claim you could not verify.

CURRENT STACK TO AUDIT (verify each license + its SaaS implication):

- ultralytics (Python lib) + the bundled YOLOv8n.pt model weights ? believed AGPL-3.0. Confirm the license of BOTH the code and the pretrained weights, whether the AGPL network clause is triggered by a closed-source hosted service that uses ultralytics server-side, and the existence/cost/terms of the Ultralytics Enterprise license.
- google-genai SDK + the Google Gemini API (model gemini-2.5-flash). Distinguish the SDK license (Apache?) from the SERVICE terms. Confirm: (a) is commercial use of outputs allowed in a paid product; (b) does Google train on / human-review prompt+video data on the PAID/billed tier vs the free tier; (c) abuse-monitoring / data-retention terms; (d) current per-token / per-video pricing for gemini-2.5-flash (input video tokens-per-second, output) so we can sketch per-video cost.
- ffmpeg / ffprobe (system binaries, invoked as separate processes). Clarify GPL vs LGPL builds and whether pure server-side SaaS use (no distribution to users) triggers any source-disclosure obligation. SEPARATELY and importantly: video CODEC PATENT royalties for a commercial service that DECODES user-uploaded H.264/AVC and H.265/HEVC (e.g. GoPro footage) and ENCODES output ? covering MPEG-LA / Via-LA (AVC), Access Advance + Via-LA + others (HEVC) pools, AAC audio (Via-LA), what activities trigger royalties, any free thresholds (e.g. the AVC sub-100k or free-internet-broadcast provisions), and whether a small SaaS realistically incurs these. Then royalty-FREE codec alternatives for the OUTPUT we generate: AV1, VP9, Opus ? and their encoder maturity/speed on a GPU VM.
- torch, torchvision ? confirm BSD; flag that some torchvision *pretrained weights* may carry separate licenses.
- opencv-python ? confirm Apache-2.0 and that the PyPI wheel's bundled components are clean for commercial server use.
- lapx (a fork of "lap" for ByteTrack association) ? confirm license.
- Quick permissive confirmation for: fastapi, uvicorn, pydantic, jinja2, python-dotenv, numpy.

PERMISSIVE ALTERNATIVES TO RESEARCH (all must be GPU-VM-friendly and usable in a closed-source

commercial SaaS):

1. Object detection to replace ultralytics/YOLOv8 for player/person detection (and possibly shuttle): compare YOLOX (Apache-2.0), RT-DETR / RT-DETRv2 (the Baidu/Apache implementations ? NOT the Ultralytics port), PP-YOLOE, MMDetection (Apache-2.0), Detectron2 (Apache-2.0), torchvision built-in detectors (Faster R-CNN/RetinaNet/FCOS/SSD, BSD), RF-DETR (Roboflow), and D-FINE/DEIM. For EACH, verify BOTH the code license AND the pretrained-weights license (some "open" detectors like YOLO-NAS have restrictive weight licenses ? confirm). Note accuracy/speed trade-offs for person detection on sports video.
2. Shuttle/ball trajectory tracking ? VERIFY the actual repo licenses (commercial-use viability) of: WASB / WASB-SBDT ("Widely Applicable Strong Baseline" for sports ball detection), TrackNetV2 and TrackNetV3, MonoTrack, and any permissively-licensed alternative for tiny-fast-object tracking. This is the planned shuttle-tracking substrate, so its commercial-use license is high-stakes.
3. LLM / semantic rally-boundary step ? research ALL THREE strategies (the user wants all three compared):
 - (a) ELIMINATE the hosted LLM entirely: permissively-licensed temporal action segmentation / video models suitable for learning rally boundaries offline (e.g. MS-TCN/MS-TCN++, ASFormer, ActionFormer, and similar) ? verify licenses and whether they fit a from-annotations learned segmenter.
 - (b) SELF-HOST an open-weights vision-language model that can ingest video frames for semantic boundaries: compare Qwen2.5-VL, InternVL2/2.5, LLaVA-OneVision/LLaVA-Video, MiniCPM-V, VideoLLaMA ? for EACH verify the WEIGHTS license for commercial use (Apache-2.0 vs custom "research-only"/"non-commercial"/community licenses with MAU caps) and the GPU VRAM needed per size tier.
 - (c) KEEP Gemini but de-risk: summarize commercial terms, data-usage on paid tier, rate limits, and per-video cost at scale, plus other hosted-API alternatives (OpenAI, Anthropic) and how their commercial/data terms compare.

DELIVERABLE: A cited report containing (1) a per-dependency RISK TABLE ? columns: dependency, license, does-SaaS-hosting-trigger-obligations?, monetization verdict (CLEAR / WATCH / BLOCKER), recommended action; (2) a recommended FULLY-PERMISSIVE target stack that eliminates all BLOCKERS; (3) paid-license fallbacks where relevant; (4) a rough per-video processing-cost sketch on a GPU VM for each of the three LLM strategies; (5) an explicit list of the codec/patent considerations and the royalty-free output-codec recommendation. Prioritize accuracy and primary-source citations over breadth."

Your angle: ****Shuttle/ball trajectory tracking repo licenses (commercial-use viability)**** ? The planned shuttle-tracking substrate lives in niche academic repos whose LICENSE files (often absent, GPL, or research-only) decide commercial viability; this high-stakes item needs explicit per-repo primary-source verification.

Search query: `WASB WASB-SBDT sports ball detection license; TrackNetV2 TrackNetV3 MonoTrack GitHub LICENSE commercial use; permissive tiny fast object tracking badminton shuttlecock trajectory`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-05-30T03:03:26.668Z)

I'll research the licenses of shuttle/ball trajectory tracking repos. Let me start by loading the web search tools.

3. user (2026-05-30T03:03:48.831Z)

Web search results for query: "WASB-SBDT widely applicable strong baseline sports ball detection tracking GitHub license"

Links: [{"title": "GitHub - nttcom/WASB-SBDT: [BMVC2023] Widely Applicable Strong Baseline for

Sports Ball Detection and Tracking · GitHub", "url": "https://github.com/nttcom/WASB-SBDT"}, {"title": "Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://arxiv.org/pdf/2311.05237"}, {"title": "[2311.05237] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://arxiv.org/abs/2311.05237"}, {"title": "Widely Applicable Strong Baseline for Sports Ball Detection and Tracking - ADS", "url": "https://ui.adsabs.harvard.edu/abs/2023arXiv231105237T/abstract"}, {"title": "Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://proceedings.bmvc2023.org/310/"}, {"title": "[2311.05237v1] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://arxiv.org/abs/2311.05237v1"}, {"title": "(PDF) Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://www.researchgate.net/publication/376208967_Widely_Applicable_Strong_Baseline_for_Sports_Ball_Detection_and_Tracking"}, {"title": "WASB-SBDT/GET_STARTED.md at main · nttcom/WASB-SBDT", "url": "https://github.com/nttcom/WASB-SBDT/blob/main/GET_STARTED.md"}, {"title": "?SBDT datasets from 5 different sport categories: - Volleyball and Basketball", "url": "https://bmvc2022.mpi-inf.mpg.de/BMVC2023/0310_poster.pdf"}]

Based on the search results, I found information about the WASB-SBDT project:

WASB-SBDT is a [BMVC2023] project on GitHub for "Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", created by nttcom.

The project presents a novel Sports Ball Detection and Tracking (SBDT) method that can be applied to various sports categories, composed of (1) high-resolution feature extraction, (2) position-aware model training, and (3) inference considering temporal consistency.

The approach was validated by comparing it with 6 state-of-the-art SBDT methods on 5 datasets from different sports categories, introducing two new SBDT datasets and providing new ball annotations for two additional datasets.

The datasets and codes are available at <https://github.com/nttcom/WASB-SBDT>.

However, the search results did not contain specific information about the GitHub license for this project. You may need to check the repository's LICENSE file directly on GitHub to find the specific license type.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

4. user (2026-05-30T03:04:01.895Z)

Software License

The repository uses the **MIT License**.

This is indicated by:

1. A license badge in the README stating "License: MIT"
2. A LICENSE.md file listed in the repository
3. The footer note: "MIT license"

The license is described as allowing "Open Source Software" usage while respecting the license terms. This is a permissive open-source license that allows broad usage with attribution.

5. user (2026-05-30T03:04:03.808Z)

Software License

License Type: MIT

The repository includes an MIT license file, as indicated in the GitHub interface where "MIT license" is referenced in the navigation. This is a permissive open-source license allowing commercial and private use with proper attribution.

****License Text Reference:**** The exact license terms are not fully displayed in the provided web content, but the MIT license typically permits use, modification, and distribution with the requirement to include the original license notice.

****Pretrained Weights:**** The documentation mentions downloadable checkpoints from Google Drive but does not specify their license terms in the provided content.

6. user (2026-05-30T03:04:12.486Z)

Web search results for query: "TrackNetV3 TrackNetV2 badminton shuttlecock tracking GitHub LICENSE commercial use"

Links: [{"title": "GitHub - qaz812345/TrackNetV3: Implementation of paper - TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification · GitHub", "url": "https://github.com/qaz812345/TrackNetV3"}, {"title": "GitHub - nickluo/TrackNetV3: TrackNetV3 modified version (TensorRT INT8 Optimization) · GitHub", "url": "https://github.com/nickluo/TrackNetV3"}, {"title": "TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification | Proceedings of the 5th ACM International Conference on Multimedia in Asia", "url": "https://dl.acm.org/doi/10.1145/3595916.3626370"}, {"title": "TrackNetV3: Enhancing ShuttleCock Tracking with ...", "url": "https://people.cs.nycu.edu.tw/~yushuen/data/TrackNetV3.pdf"}, {"title": "tracknetv3 · GitHub Topics · GitHub", "url": "https://github.com/topics/tracknetv3"}, {"title": "TrackNet-Badminton-Tracking-tensorflow2/LICENSE at main · Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2", "url": "https://github.com/Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2/blob/main/LICENSE"}, {"title": "tracknetv2 · GitHub Topics · GitHub", "url": "https://github.com/topics/tracknetv2"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "GitHub - Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2: TrackNet for badminton tracking using tensorflow2 · GitHub", "url": "https://github.com/Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2"}]

Based on the web search results, here's what I found about TrackNetV3 and TrackNetV2:

Overview

TrackNetV3 is a sophisticated model designed to enhance the precision of shuttlecock localization in broadcast badminton videos, composed of two core modules: trajectory prediction and rectification. TrackNetV2 is an efficient shuttlecock tracking network.

Key Technical Differences

The trajectory prediction module in TrackNetV3 leverages an estimated background as auxiliary data to locate the shuttlecock despite visual interferences, incorporates mixup data augmentation to strengthen robustness, and creates repair masks by analyzing predicted trajectory to rectify the path via inpainting.

Performance

TrackNetV3 yields accuracies of 97.51%, contrasted with 94.98% for TrackNetV2.

GitHub Repositories

I found references to the official TrackNetV3 implementation at GitHub (qaz812345/TrackNetV3). However, the search results did not provide specific information about the LICENSE files for TrackNetV3 or TrackNetV2 or details about commercial use restrictions. To get that specific information, you would need to visit the GitHub repositories directly and check their LICENSE files.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-05-30T03:04:18.897Z)

Web search results for query: "MonoTrack badminton monocular shuttle 3D trajectory GitHub license"

Links: [{"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | DeepAI", "url": "https://deepai.org/publication/monotrack-shuttle-trajectory-reconstruction-from-monocular-badminton-video"}, {"title": "[2204.01899] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/abs/2204.01899"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | Request PDF", "url": "https://www.researchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "(PDF) MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://www.researchgate.net/publication/359757180_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/iel7/9856930/9856648/09857202.pdf"}, {"title": "GitHub - jhwang7628/monotrack · GitHub", "url": "https://github.com/jhwang7628/monotrack"}, {"title": "Papers", "url": "https://cs.stanford.edu/people/paulliu/papers/"}, {"title": "dblp: MonoTrack: Shuttle trajectory reconstruction from monocular badminton video.", "url": "https://dblp.org/rec/journals/corr/abs-2204-01899.html"}]

Based on the search results, here's what I found about MonoTrack:

Overview

MonoTrack is an end-to-end system for reconstructing 3D and 2D trajectories from broadcast-style badminton videos. The system presents the first complete end-to-end system for the extraction and segmentation of 3D shuttle trajectories from monocular badminton videos, integrating badminton domain knowledge such as court dimension, shot placement, physical laws of motion, along with vision-based features such as player poses and shuttle tracking.

GitHub Repository

The GitHub repository is located at: <https://github.com/jhwang7628/monotrack>

However, the search results do not contain specific information about the license for the MonoTrack GitHub repository. While the search found the GitHub repository link, the license details were not included in the returned information. You may need to visit the GitHub repository directly to find the license information (typically listed in a LICENSE file in the repository root).

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8. user (2026-05-30T03:04:47.032Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

9. user (2026-05-30T03:04:50.256Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

10. user (2026-05-30T03:04:50.804Z)

License Analysis

****License Type:**** Adobe Research License

****Commercial Use:**** Not permitted

****Full License Text:****

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11. user (2026-05-30T03:04:51.642Z)

License Information for TrackNetV3

****License File Present:**** Yes, a `LICENSE` file appears in the repository's file listing.

****License Type:**** MIT License

****README Statement:**** The README does not explicitly discuss the license terms, though the MIT license file is referenced in the repository structure.

The MIT license is a permissive open-source license that allows users to freely use, modify, and distribute the software with minimal restrictions, provided they include the original license and copyright notice.

12. user (2026-05-30T03:04:56.133Z)

Web search results for query: "nttcom WASB-SBDT LICENSE custom license NOASSERTION commercial use terms"

Links: [{"title": "GitHub - nttcom/WASB-SBDT: [BMVC2023] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking · GitHub", "url": "https://github.com/nttcom/WASB-SBDT"}, {"title": "Licensing Policies and Procedure | NTT Technology Licensing Site | NTT R&D Website", "url": "https://www.rd.ntt/e/ntt-tec/procedure/"}, {"title": "WASB-SBDT/GET_STARTED.md at main · nttcom/WASB-SBDT", "url": "https://github.com/nttcom/WASB-SBDT/blob/main/GET_STARTED.md"}, {"title": "Support describing license properties and SPDX expression assertions · Issue #1577 · anchore/syft", "url": "https://github.com/anchore/syft/issues/1577"}, {"title": "NoAssertionLicense - SPDX Specification 3.0.1", "url": "https://spdx.github.io/spdx-spec/v3.0.1/model/ExpandedLicensing/Individuals/NoAssertionLicense/"}, {"title": "Open Source Licenses in SBOMs. Open Source licenses play a crucial? | by Interlynk | Medium", "url": "https://medium.com/@interlynkblog/open-source-licenses-in-sboms-9ee6f6dc1941"}, {"title": "Get your license: Transportation network company | Washington State Department of Licensing", "url": "https://dol.wa.gov/professional-licenses/transportation-network-company/get-your-license-transportation-network-company"}, {"title": "Manage licenses | Helm Docs - MedCrypt", "url": "https://helm.docs.medcrypt.com/manage-sboms/manage-sbom/manage-licenses"}, {"title": "Why should NOASSERTION be the curated license description merely because there is no SPDX identifier? · Issue #99 · clearlydefined/clearlydefined", "url": "https://github.com/clearlydefined/clearlydefined/issues/99"}]

Based on the search results, I found information about the nttcom/WASB-SBDT project and some context about NOASSERTION in licensing:

The WASB-SBDT is a Widely Applicable Strong Baseline for Sports Ball Detection and Tracking that can be applied to a wide range of sports categories.

However, the search results do not contain specific information about the custom license terms, commercial use restrictions, or the detailed licensing agreement for the WASB-SBDT project. The results indicate that NOASSERTION is used in SPDX documentation when license discovery was not attempted, or the license is intentionally not included in the document.

To find the specific licensing terms for the nttcom/WASB-SBDT project, you would need to check the LICENSE file in the GitHub repository directly or consult the project's documentation for details on commercial use restrictions and any custom license terms they may have implemented.

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13. user (2026-05-30T03:04:59.872Z)

License Analysis

License Type: Adobe Research License

Commercial Use: Not permitted

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The license explicitly prohibits commercial licensing, distribution, or any activity resulting in commercial gain. Users may reproduce, modify, and display the materials solely for academic and teaching purposes.

14. user (2026-05-30T03:05:03.121Z)

License Information

Based on the README content provided, I found the following:

****Code License:**** The repository displays a badge indicating "[![License: MIT](...)](/)" which links to the MIT license. The README does not include a full "License" section with verbatim text, but the badge clearly indicates MIT licensing.

****Commercial Use:**** The MIT license permits commercial use without restriction.

****Pretrained Weights and Dataset Licenses:**** The README does not specify the licenses for pretrained model weights or datasets. It references external documentation in `GET\_STARTED.md` and `MODEL\_ZOO.md` files, but their content is not included in the provided excerpt. Users would need to consult those files for complete licensing information on the weights and datasets.

15. user (2026-05-30T03:05:03.518Z)

License Information for MonoTrack

Based on the GitHub page content provided, I can confirm that a LICENSE file exists in the repository, but the specific license type and terms are not detailed in the page excerpt shown.

The README section does not include any explicit license discussion or commercial use guidance in the visible text.

****To determine the exact license type and commercial use permissions, you would need to:****

- View the actual LICENSE file in the repository
- Check the full README.md content
- Visit the GitHub repository directly

The page confirms a LICENSE file is present but doesn't display its contents, so I cannot identify whether it's MIT, BSD, GPL, research-only, or another license type based solely on this excerpt.

16. user (2026-05-30T03:05:07.663Z)

Web search results for query: "'WASB-SBDT' OR 'nttcom/WASB-SBDT' license MIT non-commercial 'for research'"

Links: [{"title": "(PDF) Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://www.researchgate.net/publication/376208967_Widely_Applicable_Strong_Baseline_for_Sports_Ball_Detection_and_Tracking"}, {"title": "Frequently Asked Questions | MIT Technology Licensing Office", "url": "https://tlo.mit.edu/about/frequently-asked-questions"}, {"title": "Software & Open Source Protection | MIT Technology Licensing Office", "url": "https://tlo.mit.edu/researchers-mit-community/protect/software-open-source-protection"}, {"title": "GitHub - nttcom/WASB-SBDT: [BMVC2023] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking · GitHub", "url": "https://github.com/nttcom/WASB-SBDT"}, {"title": "Research data how to license", "url": "https://www.openaire.eu/research-data-how-to-license/"}, {"title": "Licensing your data | Research | Imperial College London", "url": "https://www.imperial.ac.uk/research-and-innovation/support-for-staff/scholarly-communication/research-data-management/sharing-data/licensing-your-data/"}, {"title": "How to License Research Data | DCC", "url": "https://www.dcc.ac.uk/guidance/how-guides/license-research-data"}, {"title": "SOLIDWORKS RESEARCH LICENSE FAQ", "url": "https://www.solidworks.com/sites/default/files/2017-12/3DS-2015-SWK-EDU-WhichEditionFAQ-Flyer_0.pdf"}, {"title": "Licensing FAQs | Technology Transfer", "url": "https://www.techtransfer.nih.gov/partnerships/licensing-faqs"}]

Based on the search results, I found information about the WASB-SBDT project. The datasets and codes are available at <https://github.com/nttcom/WASB-SBDT>.

However, the search results don't contain specific information about the license type for the WASB-SBDT repository (whether it's MIT, non-commercial, or for research use). To get the exact license details, you would need to visit the GitHub repository directly at <https://github.com/nttcom/WASB-SBDT>, where the license information is typically displayed in the LICENSE file or in the README.

The search results do provide general context about different licensing types - if the intent is to release software in an open access manner without granting commercial rights to licensees/end users, a non-commercial, research use only license is suggested rather than an open source license.

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17. user (2026-05-30T03:05:10.415Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

18. user (2026-05-30T03:05:11.798Z)

Web search results for query: "qaz812345 TrackNetV3 no license all rights reserved shuttlecock"

Links: [{"title": "GitHub - qaz812345/TrackNetV3: Implementation of paper - TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification · GitHub", "url": "https://github.com/qaz812345/TrackNetV3"}, {"title": "TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification | Proceedings of the 5th ACM International Conference on Multimedia in Asia", "url": "https://dl.acm.org/doi/10.1145/3595916.3626370"}, {"title": "TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification", "url": "https://dl.acm.org/doi/fullHtml/10.1145/3595916.3626370"}, {"title": "TrackNetV3", "url": "https://www.scribd.com/document/853585931/TrackNetV3"}, {"title": "TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification - National Yang Ming Chiao Tung University Academic Hub", "url": "https://scholar.nycu.edu.tw/en/publications/tracknetv3-enhancing-shuttlecock-tracking-with-augmentations-and-/"}, {"title": "qaz812345 - Overview", "url": "https://github.com/qaz812345"}, {"title": "TrackNetV3/train.py at master · qaz812345/TrackNetV3", "url": "https://github.com/qaz812345/TrackNetV3/blob/master/train.py"}, {"title": "GitHub - nickluo/TrackNetV3: TrackNetV3 modified version (TensorRT INT8 Optimization) · GitHub", "url": "https://github.com/nickluo/TrackNetV3"}]

Based on the search results, here's what I found:

TrackNetV3 is a model composed of two core modules: trajectory prediction and rectification. The trajectory prediction module leverages an estimated background as auxiliary data to locate the shuttlecock in spite of the fluctuating visual interferences. This module also incorporates mixup data augmentation to formulate complex scenarios to strengthen the network's robustness. Given that a shuttlecock can occasionally be obstructed, repair masks are created by analyzing the predicted trajectory, subsequently rectifying the path via inpainting.

TrackNetV3 is designed to enhance the precision of shuttlecock localization in broadcast badminton videos. Experimental results illustrate a substantial enhancement over previous standard methods, increasing the accuracy from 87.72% to 97.51%.

The source code is available at <https://github.com/qaz812345/TrackNetV3>. The paper was published at ACM Multimedia Asia 2023, in December 2023 in Tainan, Taiwan.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

19. user (2026-05-30T03:05:13.975Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated

tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

20. user (2026-05-30T03:05:24.085Z)

Repository License Information

****Files in repo root:**** Yes, there is a LICENSE file listed (`LICENSE.md`).

****GitHub sidebar license indicator:**** The sidebar displays "MIT license" as the license designation.

The repository includes a `LICENSE.md` file, and GitHub identifies this project as being under the MIT license, which is an open-source software license permitting broad usage with minimal restrictions.

21. user (2026-05-30T03:05:46.259Z)

Structured output provided successfully

22. user (2026-05-30T03:07:31.825Z)

Repository License

****SPDX License ID:**** MIT

****License Name:**** MIT License

****Key Terms:**** The license permits use, copying, modification, merging, publishing, distribution, sublicensing, and selling of the software. It requires that "the above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software." The software is provided without warranties or conditions, and authors bear no liability for claims or damages arising from its use.

23. user (2026-05-30T03:07:35.544Z)

Based on the GitHub API response, the repository uses the ****MIT License****.

- ****SPDX ID:**** `MIT`
- ****License Name:**** MIT License

24. user (2026-05-30T03:07:35.569Z)

Based on the GitHub API response, this repository has:

- ****SPDX License ID:**** NOASSERTION
- ****License Name:**** Other

The repository uses a custom Adobe Research License rather than a standard open source license, which is why GitHub classifies it as "Other" with no recognized SPDX identifier.

25. user (2026-05-30T03:07:37.660Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

26. user (2026-05-30T03:07:38.575Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated

tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

27. user (2026-05-30T03:07:39.016Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

28. user (2026-05-30T03:07:39.256Z)

License Information

****SPDX ID:**** MIT

****License Name:**** MIT License

The repository uses the standard MIT License, as indicated by the GitHub API response metadata.

29. user (2026-05-30T03:07:39.462Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

30. user (2026-05-30T03:07:39.846Z)

Based on the provided GitHub repository data:

****Default Branch:**** The repository uses "main" as its default branch.

****License Information:****

- ****SPDX ID:**** "MIT"
- ****Name:**** "MIT License"

31. user (2026-05-30T03:07:39.940Z)

Adobe Research License Analysis

****License Type:**** Adobe Research License (proprietary, non-commercial)

****Commercial Use Permitted:**** No

****Header:****

"ADOBE RESEARCH LICENSE"

****Key Restriction on Commercial Use:****

"The rights granted herein may be exercised for noncommercial research purposes (i.e., academic research and teaching) only."

The license explicitly prohibits commercial activities, stating that noncommercial purposes "do not include commercial licensing or distribution, development of commercial products, or any other activity that results in commercial gain." Users may modify and redistribute the materials only under equivalent noncommercial restrictions.

32. user (2026-05-30T03:07:40.047Z)

License Analysis

****License Type:**** MIT

****First Lines:****

"MIT License

Copyright (c) 2024 alenzenx

Permission is hereby granted, free of charge, to any person obtaining a copy"

****Commercial Use:**** Yes, permitted. The MIT license explicitly allows users to "use, copy, modify, merge, publish, distribute, sublicense, and/or sell" the software without restriction, making it suitable for commercial purposes.

33. user (2026-05-30T03:07:40.255Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

34. user (2026-05-30T03:07:42.057Z)

Adobe Research License Analysis

****License Type:**** Adobe Research License

****Header/Title Quote:****

"ADOBE RESEARCH LICENSE"

****Commercial Use Permitted:**** No

****Key Restriction Quote:****

"The rights granted herein may be exercised for noncommercial research purposes (i.e., academic research and teaching) only."

****Summary:**** This license explicitly restricts use to noncommercial research and academic purposes. Commercial activities?including product development, licensing, distribution, or any activity generating commercial gain?are prohibited. Violation of these conditions automatically terminates all granted rights.

35. user (2026-05-30T03:07:45.021Z)

License Information

The README mentions a license via a badge: `[![License: MIT](https://img.shields.io/badge/License-MIT-yellow.svg)](https://opensource.org/licenses/MIT)`

This indicates the project uses an ****MIT License****.

What the README States

The README does not provide detailed sections discussing commercial use, dataset licensing, or pretrained model weights licensing. It only displays the MIT license badge without elaboration. To understand the specific terms regarding commercial use and how they apply to datasets and model weights, you would need to review the actual LICENSE file in the repository, as the README itself doesn't distinguish between these different components.

36. assistant (2026-05-30T03:07:50.434Z)

My initial findings revealed important corrections to my first-pass output: the GitHub API and raw LICENSE fetches showed that WASB-SBDT is actually MIT-licensed (not "Other"), TrackNetV3 (qaz812345) is MIT, the alenzenx fork is MIT, while MonoTrack carries the Adobe Research License (non-commercial). My earlier StructuredOutput call was based on incomplete data and got several verdicts wrong. Let me verify the key findings directly before submitting a corrected answer.

37. user (2026-05-30T03:07:55.766Z)

MIT License Analysis

****Full License Text:****

The complete MIT License text is provided above, spanning from "MIT License" through "THE SOFTWARE IS PROVIDED..." to the final clause about "THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE."

****Standard MIT License Confirmation:****

Yes, this is the standard MIT License. It contains all the required elements: permission grant, conditions regarding notice inclusion, and the liability disclaimer.

****Copyright Holder:****

"Copyright (c) 2023 NTT Communications Corporation"

****Permission Grant:****

"Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files" (120 characters)

****Commercial Use:****

Commercial use is permitted. The license grants rights to "use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software" without restriction, which explicitly includes commercial purposes.

38. user (2026-05-30T03:07:56.255Z)

MIT License Analysis

****Full License Text (Verbatim):****

"MIT License

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The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE."

****Is this the standard MIT License?****

Yes, this matches the standard MIT License template.

****Copyright Holder Line:****

"Copyright (c) 2024 qaz812345"

****Commercial Use Permitted?****

Yes. The license grants rights "to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software" without restriction, explicitly allowing commercial applications.

39. user (2026-05-30T03:07:57.123Z)

Based on the content provided, here is the license information:

****SPDX ID:**** MIT

****License Name:**** MIT License

****File Path:**** LICENSE

The license object indicates this repository uses the MIT License, which permits broad usage rights including modification and distribution, provided the original copyright notice is retained.

40. user (2026-05-30T03:07:57.937Z)

License Analysis

****License Type:**** Yes, this is an MIT License.

****Copyright Line:****

"Copyright (c) 2024 qaz812345"

****Permission Grant:****

"Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software"

****Commercial Use:**** Yes, commercial use is explicitly permitted. The license grants rights to modify and distribute the software for any purpose, including commercial ones, provided the copyright notice and license terms are included in all copies or substantial portions of the software.

41. user (2026-05-30T03:07:59.792Z)

Repository Information

****Full Name:**** nttcom/WASB-SBDT

****Description:**** "[BMVC2023] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking"

****License:****

- ****SPDX ID:**** MIT

- ****Name:**** MIT License

42. user (2026-05-30T03:08:01.053Z)

License Analysis

****License Type:**** MIT License

****Copyright Holder:**** The page states "Copyright (c) 2023 NTT Communications Corporation"

****Permission Grant:**** "Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software"

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****Commercial Use?**** Yes, commercial use is permitted. The license explicitly grants rights to "use, copy, modify, merge, publish, distribute, sublicense, and/or sell" the software without restriction, which includes commercial purposes.

43. user (2026-05-30T03:08:05.795Z)

License Information

Based on the webpage content provided:

****License Indicator Text:**** The right sidebar displays "[MIT license](#MIT-1-ov-file)" as a clickable link.

****LICENSE File:**** Yes, there is a LICENSE file present in the repository file listing, appearing in the files and folders section near the top of the repository.

The repository uses the MIT license, which is a permissive open-source license that allows broad reuse of the code.

44. user (2026-05-30T03:08:07.218Z)

License and Citation Analysis

The provided README excerpt ****does not mention any license, citation requirement, or commercial-use restriction****.

Key Observations:

****License Information:**** Absent from the content shown.

****Citation:**** The README references an academic paper: "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video" (CVPR 2022), but does not explicitly request citations in the README itself.

****Dependencies with Potential Licenses:****

- The system relies on ****MMPose**** and ****MMDet**** (external dependencies with their own licensing terms)
- A ****modified TrackNet**** is used for shuttle tracking, though the README doesn't specify the original TrackNet version's license details
- The excerpt references TrackNetV2 indirectly through the modified implementation but provides no license information

****Recommendation:**** Users should consult the individual dependency repositories (MMPose, MMDet, TrackNet) and the full project repository for complete license information, as the excerpt provided lacks these details.

45. user (2026-05-30T03:08:23.419Z)

Structured output provided successfully

46. assistant (2026-05-30T03:08:33.944Z)

All findings are now verified against primary sources. The corrected, final structured output has been submitted. Summary of the verified license findings for shuttle/ball trajectory tracking substrate:

- nttcom/WASB-SBDT (BMVC 2023): MIT License, "Copyright (c) 2023 NTT Communications Corporation" ? confirmed via GitHub API (spdx_id=MIT) and raw LICENSE.md. CLEAR for commercial SaaS; watch dataset/weights terms separately.
- qaz812345/TrackNetV3 (ACM MM Asia 2023, the official repo): MIT License, "Copyright (c) 2024 qaz812345", LICENSE on the `master` branch ? confirmed via API and raw file. CLEAR; Google Drive checkpoints carry no explicit grant (WATCH).

- alenzenx/TrackNetV3: MIT License, "Copyright (c) 2024 alenzenx" ? confirmed via raw LICENSE. CLEAR.
- jhwang7628/monotrack (CVPRW 2022): Adobe Research License ? noncommercial research/teaching only, "any other activity that results in commercial gain" prohibited, auto-termination on breach. BLOCKER for a paid SaaS; also depends on a modified TrackNet + MMPose/MMDet.
- Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2: has an explicit LICENSE file (TrackNet forks are typically MIT) ? a medium-relevance additional permissive option to confirm before adoption.

Note: my first StructuredOutput call contained errors (it reported WASB/TrackNetV3 as "Other"/no-license based on transient 404s and main-vs-master branch mismatches); the final call corrects all verdicts against verified primary sources. The structured output has been delivered.